

Slides adapted from B. Bert Gertsman, other sources are referenced on the slides

CS4048 — DATA SCIENCE

Lecture 17

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FAST – NUCES, CFD Campus

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TODAY'S TOPICS

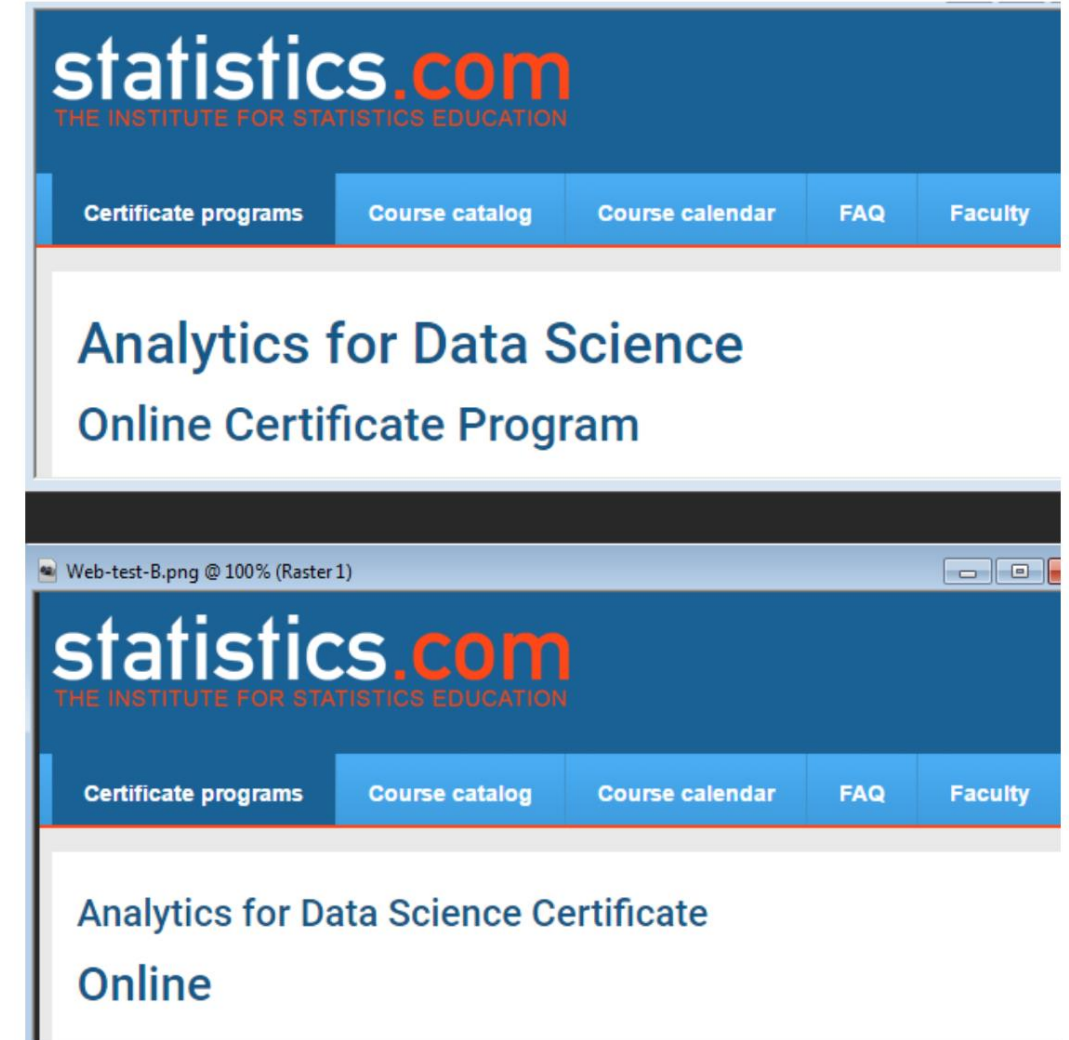
- A/B testing
- Bootstrap sampling
- Correlation vs. Covariance

A/B TESTING

- An A/B test is an experiment with two groups to establish which of two treatments, products, procedures, or the like is superior.
- Often one of the two treatments is the standard existing treatment, or no treatment.
- If a standard (or no) treatment is used, it is called the **control**.
- A typical hypothesis is that a new treatment is better than the control.
- A/B testing is a basic randomized control experiment and is very popular in data science.

A/B TESTING EXAMPLE

Testing two web headlines to determine which produces more clicks



TERMINOLOGIES

- **Treatment:** something (drug, price, web headline) to which a subject is exposed.
- **Treatment group:** a group of subjects exposed to a specific treatment.
- **Control group:** a group of subjects exposed to no (or standard) treatment.
- **Randomization:** The process of randomly assigning subjects to treatments.
- **Subjects:** The items (web visitors, patients, etc.) that are exposed to treatments.
- **Test statistic:** the metric used to measure the effect of the treatment.

WHY A/B TESTING?

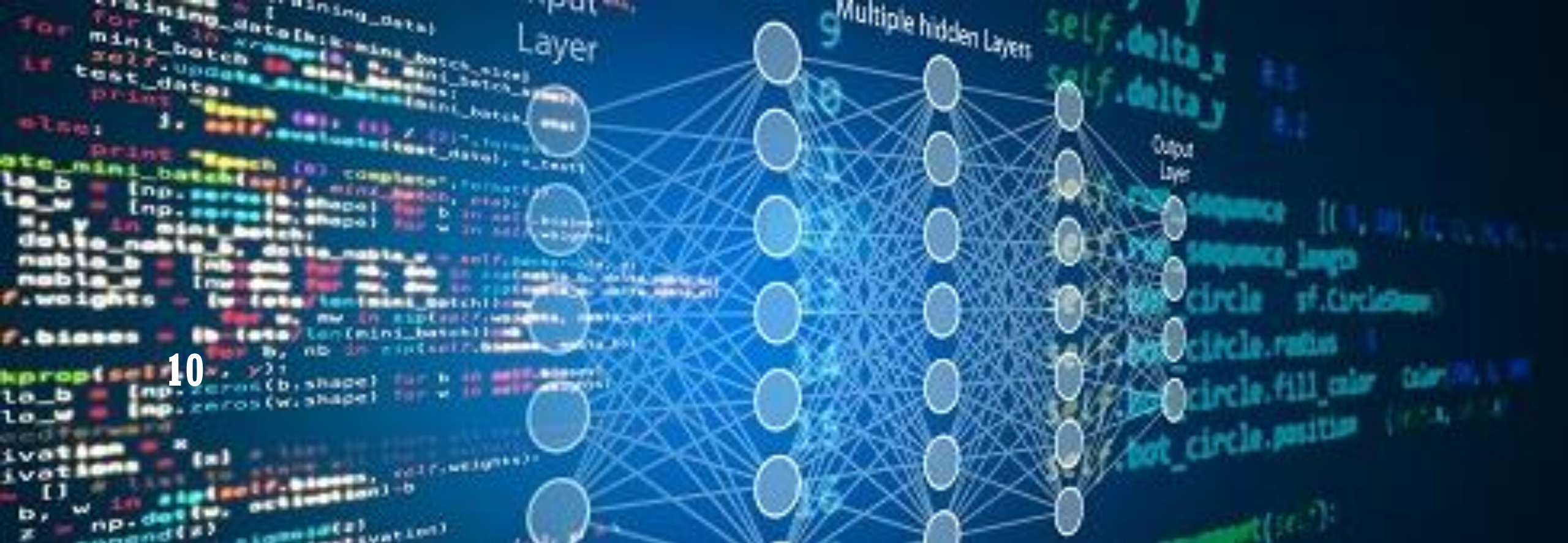
- You have made certain changes to your website recently. Unfortunately, you have no way of knowing with full accuracy how the next 100,000 people who visit your website will behave. That is the information we cannot know today, and if we were to wait until those 100,000 people visited our site, it would be too late to optimize their experience.
- A data scientist can perform A/B testing for sample data instead and perform hypothesis testing.

WHY A/B TESTING?

- Perform A/B test on a sample of 1000 first users who will visit the website
- Formulate hypotheses
- Randomly assign users to control and treatment group
 - Control group: users who will visit the old website
 - Treatment group: users who will visit the new website

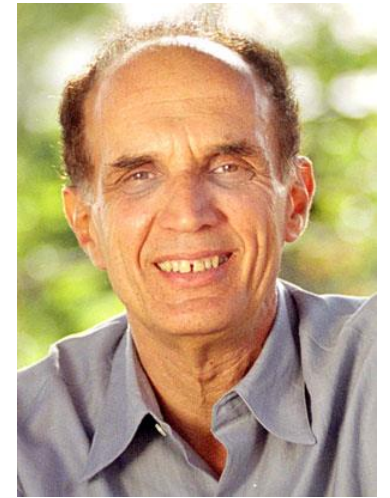
A/B TESTING

- Split your (sample) target audience or user base into two random groups.
- Show each group a different version (A or B) of the element you're testing.
 - This could be a website layout, email format, product pricing, advertisement, etc.
- Collect data on user behavior and measure each version's impact on a specific metric (e.g., click-through rate, conversion rate, sales).
- Analyze the data statistically to determine if a performance difference exists between A and B.



BOOTSTRAP SAMPLING

BOOTSTRAP SAMPLING



- Described by Bradley Efron (Stanford) in 1979.
- **Bootstrapping** refers to the ability to **estimate parameters based solely on one sample of data from the population** without any other assumptions.
 - A non-parametric method
- The bootstrap uses computer simulation.
- The bootstrap:
 - draws observations only from your own sample (not a hypothetical world)
 - makes no assumptions about the underlying distribution in the population

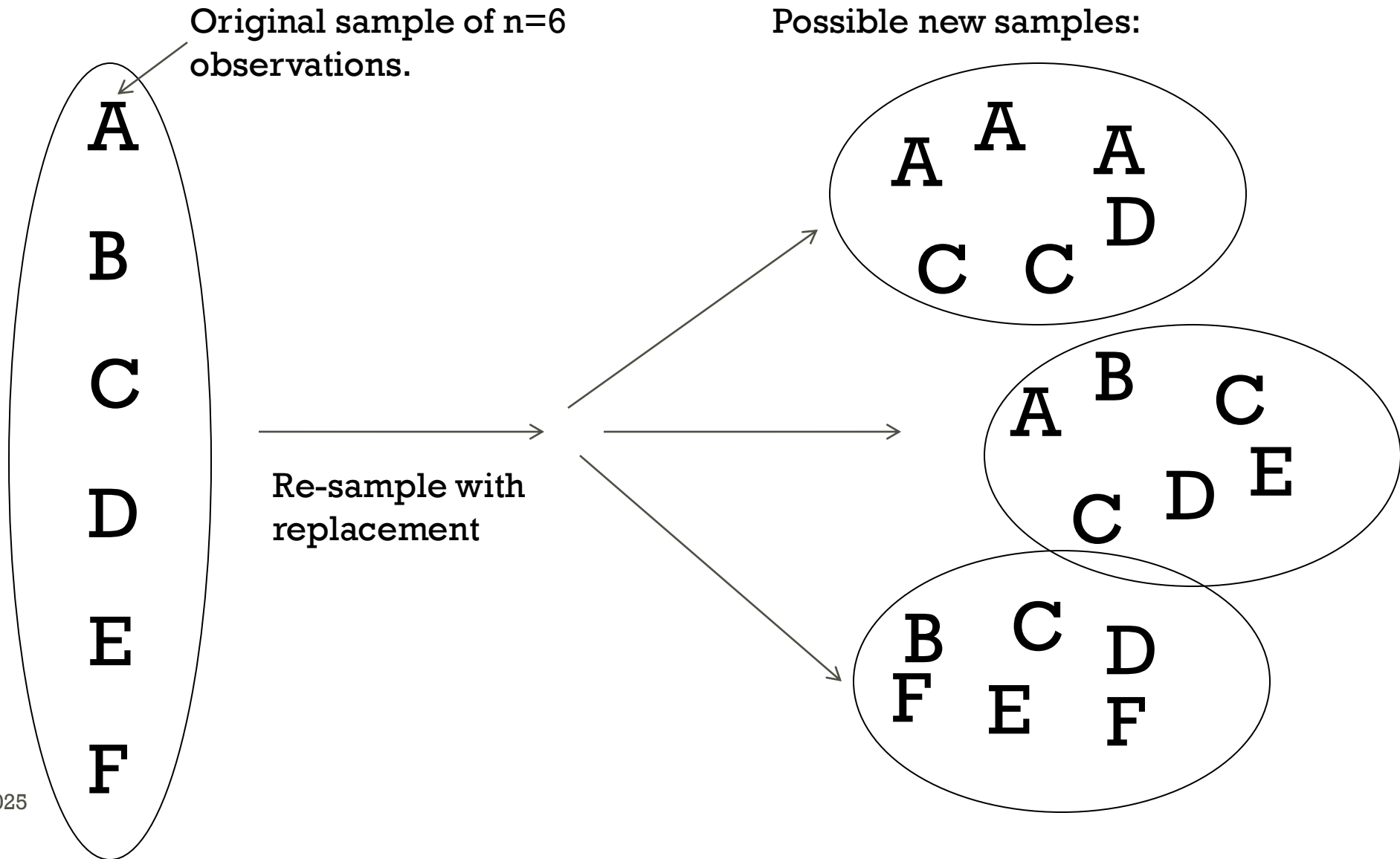
WHY BOOTSTRAP?

- Classical tests (e.g. t-test, z-test) assume that the data is distributed in a certain way.
- Allows you to estimate parameters (standard error, confidence intervals) when no formulas are available.
- Allows you to estimate parameters (standard error, confidence intervals) when assumptions are not met (e.g., large sample, normality)

SAMPLING WITH REPLACEMENT

- Sampling with replacement means every observation has an equal chance of being selected ($=1/n$), and observations can be selected more than once.

SAMPLING WITH REPLACEMENT



BOOTSTRAP PROCEDURE

1. Number your observations 1,2,3,...n
2. Draw a random sample of size n WITH REPLACEMENT.
3. Calculate your statistic (mean, median, accuracy, beta coefficient, ratio, etc.) for each bootstrap sample.
4. Repeat steps 1-3 many times (e.g., 1000 resamples).
5. Once we have a bootstrap sampling distribution there are two methods for constructing a confidence interval:
 - a. Calculate the variance of your statistic directly from your sample of 1000 statistics (for standard error).
$$\text{sample statistic} \pm 2(\text{standard error})$$
 - b. You can also calculate confidence intervals directly from your sample of 1000 statistics. Where do 95% of statistics fall?

WHEN IS BOOTSTRAP USED?

- If you have a new-fangled statistic without a known formula for standard error.
 - e.g. male: female ratio.
- If you are not sure if large sample assumptions are met.
 - Maximum likelihood estimation assumes “large enough” sample.
- If you are not sure if normality assumptions are met.
 - Bootstrap makes no assumptions about the distribution of the variables in the underlying population.

THE PROCEDURE IS QUITE STRAIGHTFORWARD

- To obtain a sense of how stable a sample statistic is (how it distributes).
- Simply **resample** (with replacement!) from the existing sample.
- Do this n times ($10k \rightarrow 100k \rightarrow 1m$)
- Calculate the standard deviation of these resampled means.
- Compare with the figure of interest.

Example covered from

BOOTSTRAP DRAWBACK/ASSUMPTION

- This works if and only if the sample truly is representative of the population.
- In other words, if something didn't happen in the sample, it can't ever happen.
- “The data we have is the only data that can ever be”
- So small samples necessarily over- or underestimate the probability of rare events.

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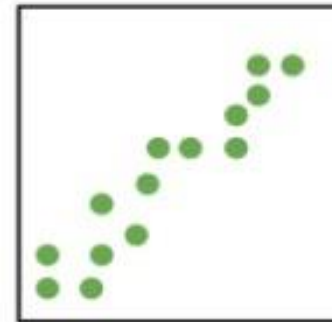
CORRELATION VS. COVARIANCE

COVARIANCE AND CORRELATION

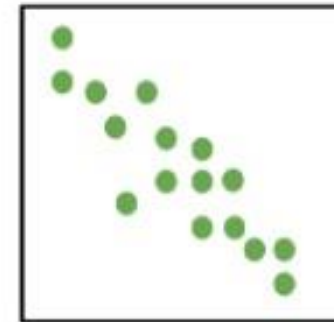
- Statistical methods to measure relationship between two numerical attributes
- A relationship exists when changes in one variable tend to be accompanied by consistent and predictable changes in the other variable.
- **Covariance** and **Correlation** coefficient (also called **Pearson's product moment coefficient**) can be used for this purpose.
 - Covariance: measures how the two variables differ
 - Correlation: measures how the two variable are related

COVARIANCE

- Covariance is a measure to determine how two attributes change together (i.e., similar or opposite directions?)
- A covariance can be positive, negative, or zero
 - A **positive covariance** means both are directly related.
 - A **negative covariance** indicates an indirect relation.
 - A **zero covariance** tells both variables are independent of the changes in each other.



Positive
Covariance



Negative
Covariance



Zero
Covariance

COVARIANCE

- Consider two numeric attributes, A and B, and a set of n observations $\{(a_1, b_1), \dots, (a_n, b_n)\}$.
- Mean values of A and B (known as expected values) are computed as:

$$E(A) = \bar{A} = \frac{\sum_{i=1}^n a_i}{n} \qquad E(B) = \bar{B} = \frac{\sum_{i=1}^n b_i}{n}.$$

- Covariance between attributes A & B is defined as:

$$\text{Cov}(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}.$$

- Alternatively, we can write it as:

$$\text{Cov}(A, B) = E(A \cdot B) - \bar{A}\bar{B}.$$

EXAMPLE

Stock Prices for *AllElectronics* and *HighTech*

Time point	<i>AllElectronics</i>	<i>HighTech</i>
t1	6	20
t2	5	10
t3	4	14
t4	3	5
t5	2	5

If the stock prices are affected by the same industry trends, will their prices rise or fall together?

and

$$E(\text{AllElectronics}) = \frac{6 + 5 + 4 + 3 + 2}{5} = \frac{20}{5} = \$4$$

$$E(\text{HighTech}) = \frac{20 + 10 + 14 + 5 + 5}{5} = \frac{54}{5} = \$10.80.$$

$$\text{Cov}(A, B) = E(A \cdot B) - \bar{A}\bar{B}.$$

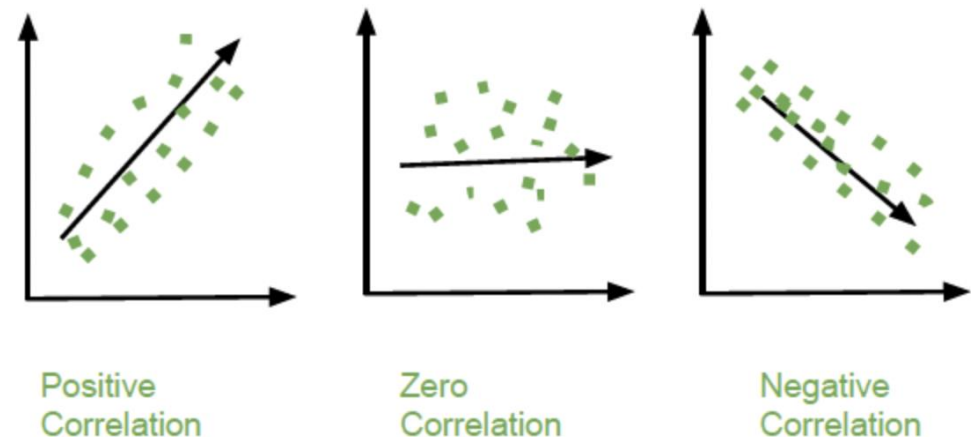
$$\begin{aligned}\text{Cov}(\text{AllElectronics}, \text{HighTech}) &= \frac{6 \times 20 + 5 \times 10 + 4 \times 14 + 3 \times 5 + 2 \times 5}{5} - 4 \times 10.80 \\ &= 50.2 - 43.2 = 7.\end{aligned}$$

Given the positive covariance, we can say that stock prices for both companies rise together.

Any issue with Covariance?

CORRELATION

- Correlation is a standardized measure of the strength and direction of the **linear relationship** between two variables.
- It is derived from covariance and ranges between -1 and 1.



PEARSON'S CORRELATION COEFFICIENT

The **Pearson correlation** measures the direction and degree of linear (straight line) relationship between two variables.

$$\text{corr}(\mathbf{x}, \mathbf{y}) = \frac{\text{covariance}(\mathbf{x}, \mathbf{y})}{\text{standard_deviation}(\mathbf{x}) * \text{standard_deviation}(\mathbf{y})} = \frac{s_{xy}}{s_x s_y}, \quad (2.11)$$

where we are using the following standard statistical notation and definitions

$$\text{covariance}(\mathbf{x}, \mathbf{y}) = s_{xy} = \frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x})(y_k - \bar{y}) \quad (2.12)$$

$$\text{standard_deviation}(\mathbf{x}) = s_x = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (x_k - \bar{x})^2}$$

$$\text{standard_deviation}(\mathbf{y}) = s_y = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (y_k - \bar{y})^2}$$

$$\bar{x} = \frac{1}{n} \sum_{k=1}^n x_k \text{ is the mean of } \mathbf{x}$$

$$\bar{y} = \frac{1}{n} \sum_{k=1}^n y_k \text{ is the mean of } \mathbf{y}$$

Correlation is always in the range of -1 to 1 (representing perfect negative and positive linear relationship).

CORRELATION (VIEWED AS LINEAR RELATIONSHIP)

- Correlation measures the linear relationship between objects
- To compute correlation, we standardize data objects, A and B, and then take their dot product

$$a'_k = (a_k - \text{mean}(A)) / \text{std}(A)$$

$$b'_k = (b_k - \text{mean}(B)) / \text{std}(B)$$

$$\text{correlation}(A, B) = A' \bullet B'$$

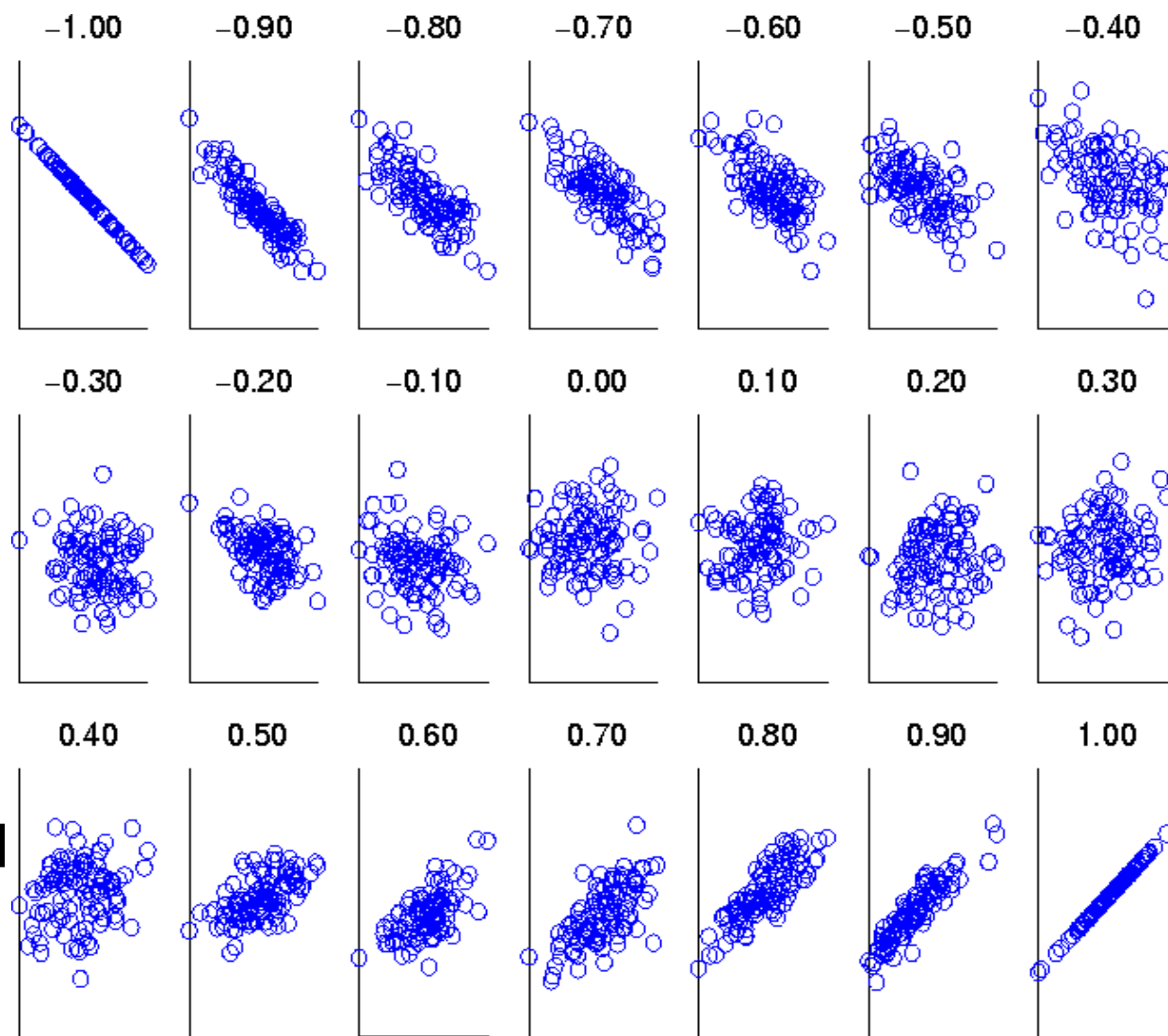
VISUAL REPRESENTATION OF CORRELATION

If $\text{corr}(x,y) > 0$, x and y are positively correlated (i.e., x 's values increase as y 's). The higher, the stronger correlation.

$\text{corr}(x,y) = 0$: both are independent

$\text{corr}(x,y) < 0$: negatively correlated

Scatter plots showing the similarity from -1 to 1 .



**CORRELATION DOES NOT
IMPLY CAUSALITY!**

See **Spurious Correlations** (<https://www.tylervigen.com/spurious-correlations>)

READING MATERIAL

- Chapter# 4: Principle of Data Science
- Chapter# 3: Practical Statistics [Bruce]
- Inferential statistics + code: <https://brainalystacademy.com/inferential-statistics-in-python/>
- A useful resoure on inferential statistics:
<https://www.statisticsfromatoz.com/blog/category/new-video/4>
- An article: Confidence Intervals for Machine Learning
(<https://machinelearningmastery.com/confidence-intervals-for-machine-learning/>)
- Notes for significance testing
(<https://home.csulb.edu/~msaintg/ppa696/696stsig.htm>)